Fiscal Year:	FY 2005	**Task Last Updated:**	FY 05/08/2006
PI Name:	Inverardi, Luca M.D.		
Project Title:	Cell transplantation therapy for diabetes utilizing immunoprivileged Sertoli-Islet cell aggregates (SICA)		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Cellular biotechnology & tissue engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	linverar@med.miami.edu	**Fax:**	FY 305-243-4404
PI Organization Type:	UNIVERSITY	**Phone:**	305-243-5347
Organization Name:	Diabetes Research Institute		
PI Address 1:	University of Miami School of Medicine		
PI Address 2:	1450 NW 10th AVE		
PI Web Page:			
City:	Miami	**State:**	FL
Zip Code:	33136	**Congressional District:**	17
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-03
Start Date:	08/01/2001	**End Date:**	03/31/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Kirby, Mary	**Contact Phone:**	
Contact Email:	mary.f.kirby@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Cameron, Don (University of South Florida)		
Grant/Contract No.:	NAG9-1346		
Performance Goal No.:			
Performance Goal Text:			

Type 1 insulin-dependent diabetes mellitus is a dysmetabolic disease caused by the destruction of insulin-producing beta cells in the pancreas by an autoimmune process. Type 1 diabetes, if left untreated, is a deadly disease. Current therapy is based on the administration of exogenous insulin, and it is quite often characterized by the development of complications, including micro- and macro-angiopathy, neuropathy, blindness and kidney failure. Biological replacement of insulin-producing tissue via transplantation of islets of Langerhans represents a palatable therapeutic option, since it is characterized by excellent metabolic control, and it is likely to prevent complication occurrence or halt their progression. Success of recent clinical trials of islet transplantation has re-ignited the scientific community's enthusiasm for this procedure, but has also underlined the remaining problems that characterize this therapeutic approach. In particular, there is a need for chronic immunosuppression of the recipient to prevent allorejection and recurrence of autoimmunity. This limits the use of islet transplantation to a small cohort of patients where the risks connected to immunosuppression are justified by untreatable diabetes. Strategies that might allow the implementation of

Task Description:	successful transplantation protocols without the need for chronic immunosuppression would revolutionize this field making this therapeutic option available to many more diabetic patients. A defined population of cells found in the testis, named Sertoli cells, has recently shown remarkable features that incude scarce immunogenicity and protection of co-transplanted tissues from immune rejection. These features make Sertoli cells potentially instrumental in promoting survival of co-transplanted islets of Langerhans in the absence of systemic immunosuppression. Indeed encouraging preliminary data has shown that Sertoli cell-islet mixtures could be successfully co-transplanted in immunocompetent mice and survive long-term. The availability of microgravity culture conditions, through the use of NASA rotating culture vessels, allows us to test the hypothesis that three-dimensional aggregates of Sertoli cells and islets will provide sizable advantages in terms of islet function and reduced immunogenicity. Rodents will be utilized as recipients of SICA (Sertoli cells-islet aggregates) to obtain proof of principle that this hypothesis is indeed correct. Syngeneic recipients, as well as allogeneic recipients will be utilized to assess function and immunoprivilege. Autoimmune diabetic rodents will also be utilized to obtain information of the role of Sertoli cells in preventing recurrence of islet destruction by autoimmune effector cells. Islets and Sertoli cells from a different donor species (pig) will also be tested, for survival and function. Results will have potential high impact on islet transplantation strategies in humans. It is the central hypothesis of our research that Sertoli cell-islet aggregates (SICA) obtained in rotating wall vessels (RWV) will engraft and function long-term, when transplanted in immunocompetent hosts. To assess this hypothesis we have proposed the following Specific Aims: 1) To assess the functional competence of SICA by transplantation in syngeneyc recipients. 2) To assess SICA immunoprivilege by transplantation in allogeneic recipients 3) To assess SICA competence in preventing recurrence of islet cell autoimmune destruction by transplantation in autoimmune-diabetic rats 4) To analyze the protective effect of Sertoli cells on islets when both allorejection and autoimmunity concur to the destruction of the graft by transplantation of allogeneic SICA in autoimmune rats 5) To evaluate competence of xenogeneic SICA by transplantation of pig Sertoli cell-islet aggregates in rats. Data obtained in these experimental settings will allow to obtain an in-depth evaluation of the therapeutic strategies of transplantation of immunoprivileged, insulin-producing tissue for the treatment of diabetes in the clinical setting. Type 1 diabetes mellitus is a disease characterized by the common occurrence of dreadful complications. It is also a disease that accounts for monumental health care expenditures. Success in biologic replacement of insulin-producing tissue by transplantation is a desirable goal. While the recent results of several clinical trials have indeed provided proof of principle that islet transplantation can work, the need for chronic immunosuppression of the recipients renders this procedure available to a small, selected group of patients. The use of Sertoli cell-islet agggregates obtained by culture in rotating wall vessels represents a potentially promising approach to solve this critically important problem. The potential impact of this research on the quality of life of Type 1 diabetic patients is unfathomable.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The benefit of this research could encompass the definition of safe, benign therapeutic means to treat diabetes mellitus, one of the most important causes of morbidity and mortality in the general human population.
Task Progress:	Recent experiments performed utilizing new rotating vessels with modified engineering showed that islets cultured under these experimental conditions have largely preserved viability and respond to glucose stimulation by secreting insulin. While small aggregates are still formed, they can be relatively easily dissociated, unlike the ones produced in old-generation rotors (vessels). We have recently developed a quite sophisticated method to assess islet beta cell-specific viability, based on the analysis of mitochondrial membrane potential (a marker of apoptosis) in the beta cell subset as identified by its high zinc content. This will allow us to assess the value of the recently described modification in the media composition in a quite reliable manner. Therefore, we are confident that we will be able to evaluate the promise of the use of new rotors and new culture conditions in achieving some of the goals stated at the inception of this funding period.
Bibliography Type:	Description: (Last Updated: 06/02/2006)
Articles in Peer-reviewed Journals	Luca G, Calvitti M, Nastruzzi C, Macchiarulo G, Becchetti E, Neri LM, Capitani S, Basta G, Brunetti P, Calafiore R, Cameron DF. "Effects of simulated microgravity on the morphology and function of neonatal porcine cell clusters cultured with and without Sertoli cells." Cell Transplant. 2006;15(1):55-65. PMID: 16700330 , Apr-2006